Customer Churn Prediction for Gyms / Coaching Centers

A Machine Learning Approach

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1. Abstract

Customer retention is crucial to business success in subscription services such as gyms and coaching facilities where steady revenue streams rely on retaining clients. A key challenge to having a steady revenue stream from these types of businesses is customer churn, or when customers cancel their memberships, do not participate in the service anymore, or do not renew their membership at the end of their subscription term. In the case of gyms, client loss has a much less distinct beginning since typically it gradually occurs over time due to reduced attendance, lack of engagement, and inconsistent payment behaviours, vs a sudden loss of a client due to their immediate cancellation of their membership. Realising the loss of a client after they have already left indicates a missed opportunity to retain this client,which contributes to ongoing revenue loss.The project aims to develop a machine learning-based system to predict client churn. By finding potential customers before they churn, businesses can improve client satisfaction and business sustainability through targeted client retention strategies, such as client engagement initiatives or tailored promotional offers.In the current study, the Gym Customers Features and Churn Dataset was used to train different classification models. The dataset is a supervised classification dataset obtained from Kaggle consisting of client demographics, length of membership, frequency of attendance, payment behaviours, and labels identifying whether a particular client has churned or not. The target variable indicates whether or not a customer has left the service provider's business, providing the parameters required to train multiple classification models using the preprocessed dataset.The research evaluates three machine learning models (Random Forest Classifier, XGBoost Classifier, and Logistic Regression) that use the latter, as it is simple to use and interpretable, as the baseline. Random Forest is used to determine the non-linear relationships within the data set by using ensemble(s) of decision trees. XGBoost is used due to its excellent predictive performance and scalability when working with large datasets utilizing the gradient-boosting technique.In evaluating the performance of the models through the use of several performance indicators (accuracy, misclassification rate, precision, recall, true positive rate, false positive rate, area under the receiver operator curve (AUC), training loss, and testing loss) and through the use of ROC curve analysis, the models can be compared with regard to their ability to discriminate between the differentclasses.The results of the study identified the best model for prediction of churn to be XGBoost, exhibiting the lowest misclassification rate, highest accuracy, recall, and AUC score. Although Logistic Regression provided good interpretability, it scored low on recall; whereas, Random Forest performed well overall .Therefore, XGBoost was the most accurate and reliable model for churn prediction .The study indicates how

machine learning models - particularly ensemble approaches - can be useful for proactively predicting customer attrition. The performance of XGBoost is superior to the performance of the other models; it provides the highest accuracy, recall, and AUC scores as well as the lowest misclassification rate. Random Forest also performs well but has the highest test loss while Logistic Regression has relatively low recall but provides good interpretability. XGBoost has been identified as the best choice for modelling the probability of a customer churning based on its overall performance.Based on what was observed in this study, machine learning models and the ensemble techniques specifically, will provide gymnasiums and coaching centres with an effective means of proactively predicting customer churn to reduce churn, improve retention strategies, and ultimately reduce lost revenue.

1. Introduction

1.1 Application Area

According to the comparative results, XGBoost is superior to all other models, with the best performance on all three metrics (highest accuracy, recall, and area under curve) while suffering from the least amount of misclassifications. Random Forest has good performance, but higher test loss than XGBoost; whereas Logistic Regression provides reasonable interpretability but relatively lower recall compared to XGBoost. Overall, it appears that the best choice for a model predicting customer churn is XGBoost. This research has shown that machine learning models, particularly ensemble-type models, can be effective at predicting customer churn before they make a cancellation. The proposed system will aid gyms and coaching facilities in reducing customer churn, enhancing retention efforts, and decreasing revenue attrition by helping identify potential loss of customers through predictive analytics. Normally gym memberships and coaching facilities utilize both a subscription business model and renewals of their memberships on an ongoing basis in order to support continuing fixed income from subscriptions while promoting future growth.Gym, coaching facility, and similar businesses rely more on retention than on single sale-type businesses and, consequently, must use the right methods of predicting churn to accomplish their operational goals. Typically, customers do not cancel their memberships suddenly; rather, membership cancellations usually occur because the customer stops visiting/participating in the gym or other coaching activities or cumulatively reduces the amount of money they spend on "ancillary" services or postpones their membership renewal (which occurs at least every month) and on average after 1-2 months of not participating or spending money. These behavioral changes typically occur gradually, over time, and as a result are usually missed by businesses, leading to a loss of customers whom the company did not expect to lose.AI

techniques offer a data-focused way to reveal patterns and trends that aren't obvious. As an example of how thoughtful creation of a churn prediction model can provide actionable insights, gyms and/or coaching facilities can create and utilize personalized offers, targeted communications, and enhance interaction with the customer base as a way of increasing the customer's loyalty, which could reduce the loss of revenue, to the gym/coaching facility.

1.2 Why Machine Learning Models Are Chosen

To find customers who may leave (churn) is possible by utilizing many different kinds of techniques. The traditional techniques include manual observation, customer feedback surveys, and statistical analysis of available data. These techniques are often reactive to loss and are not particularly useful. In addition to being time-consuming and prone to human factors, they can struggle to process very large datasets. Due to their inability to capture complex, non-linear relationships between different factors pertaining to customersit leaves the majority of these methods are unable to effectively predict .The use of large volumes of data to identify patterns and relationships that are not readily identified through traditional methods, machine learning models overcome these limitations. They can provide accurate and scalable predictions, manage a large amount of high-dimensional data, it can adjust to changes in the customers behaviour. Three separate models are selected for this project to balance considerations of interpretability,robustness and prediction accuracy.The baseline model used is logistic regression due to its straightforwardness and capacity for simple interpretation giving a clear idea impact customer churn. Random Forest also works as the advanced model since it is a general-purpose learning algorithm and uses several decision trees together to enhance performance by reducing prediction error overfitting. XGBoost, on the other hand, has been selected for use as an advanced model because it ranks well when applied to structured data, is able to effectively deal with missing data, includes methods for regularization, and has a proven capacity to identify complex interactions between variables. In essence, the combination of logistic regression, Random Forest and XGBoost provides an excellent framework for studying the phenomenon of customer attrition, with both high forecast accuracy and interpretability assured.

3. Literature Survey

Reference's [1], Customer Churn Dataset contained variables related to behaviors (such as service usage and subscription history) and demographics; these were analyzed in order to predict customer attrition using a Random Forest and Logistic

Regression models. Random Forest outperformed Logistic Regression due to its ability to capture more complex, non-linear relationships/relationships among the independent variables than Logistic Regression could. As such, Logistic Regression was unable to effectively predict complex customer churn patterns extensively analyzed. However, the study did not analyze more complex boosting methods (e.g. XGBoost and Gradient Boosting) that might have provided additional improvements to prediction performance, as well as additional robustness to models' predictions. Reference's [2] dataset contained indicators of ISP customer usage (e.g. account usage patterns) as well as customer bills and payment patterns; therefore, the Random Forest and Decision Tree algorithms were utilized to identify patterns of customer churn. A telecommunications industry customer turnover database containing rich detail about service usage and customer activity was analyzed in Reference [3]. A weighted ensemble learning approach using multiple machine learning classifiers was proposed to improve prediction accuracy and stability. The ensemble model outperformed individual models with respect to generalization and variance reduction. However, there are challenges to the use of this ensemble framework in real-time applications and business decision-making in which transparency is often critical due to the increased computational complexity and reduced interpretability of the ensemble framework. Reference [4] studied churn prediction using gradient boosting algorithms (i.e., catboost) on structured customer data based on demographic and service usage characteristics. The findings supported the idea that boosting algorithms are effective at handling categorical variables and improving prediction accuracy when used with structured data. Although the results were promising, there was no direct comparison to xgboost—one of the commonly used gradient boosting algorithms.The application of XGBoost, as mentioned by reference [5], was conducted on benchmark structured datasets where the high prediction accuracy, scalability, and speed were proved to be true for the XGBoost algorithm. Some key strengths were also emphasized by reference [5], such as the use of regularization, missing value handling, and parallel processing. Nonetheless, the testing of the algorithm in reference [5] was done by using general benchmark datasets; thus, the algorithm's performance in predicting customer churn is yet to be tested on a more specific case. Through the use of a real-world customer churn dataset which contained transactional and behavioral factors, the performance of the XGBoost algorithm in modeling interaction between factors was proven by reference [6].generalizable results across multiple data splits. The study did state, however, that rigorous tuning of hyperparameters and domain knowledge are required in order to achieve optimal results; therefore the results were not as good as those presented by reference [5].Reference [6] made use of XGBoost in predicting churn from real-life customer data having behavioral and transactional

features. The result obtained indicated that there were high accuracy and robustness due to its ability to learn complex feature interaction and nonlinearities effectively. Furthermore, it was observed that XGBoost had great generalizability in terms of accuracy with different splits of the data set. Nevertheless, it was observed that to have the best results, hyperparameter optimization and domain knowledge are important.Reference [7] showed Random Forest as an effective and reliable algorithm for ensemble learning that prevents overfitting using bagging and randomness of feature selection by conducting experiments on various benchmark data sets. Ability to work with high dimensional data without any loss in predictive accuracy was highlighted in the research paper. Nonetheless, Random Forest is computationally expensive and difficult to interpret in contrast to other simple algorithms.

4. Dataset Description

The project uses the Gym Customers Features and Churn Dataset obtained from Kaggle. It is a supervised classification dataset containing customer behavior and demographic information.

Dataset Link : (https://www.kaggle.com/datasets/adrianvinueza/gym-customers-features-and-churn)

Dataset Characteristics:

• Membership duration

• Attendance frequency

• Payment behavior

• Customer demographics

Target Variable:

• Churn (0 – No, 1 – Yes)

The dataset is well-structured and labeled, making it suitable for supervised learning and comparative model evaluation.

5. Model Description

Description of how each learning algorithm works in the proposed churn prediction system

5.1 Logistic Regression

Working in the Proposed System

The logistic regression algorithm is used as a baseline classifier for predicting churn within the proposed churn prediction system. The characteristics of clients like tenure, attendance frequency, and billing behavior to estimate the probability of clients churning takes by algorithm. The algorithm computes a weighted sum of numeric vector features of customers. The output is then converted into a probability score ranging from 0 to 1 using the sigmoid function. Clients with scores higher than the predetermined threshold are classified as churners.

The project goal got achieved through that.

Objective Solved in the Project

The goal here with logistic regression is mostly about setting up a simple baseline model to predict churn**.** You know, like figuring out if customers are going to leave or not based on data. It also helps understand how customer factors influence that in a linear way, sort of straight forward effects on behavior. I am not totally sure if linear is the best word for it, but it seems to capture the direct impacts. Anyway, thats the main idea behind using it for this.

Justification for Model Selection

Logistic regression is basically for building a simple baseline model to predict churn. You know, looking at customer features and seeing how they affect behavior with this linear setup. It feels like that's one of the main goals, anyway.

Advantages:

• Serves as a benchmark for assessing performance.

• Shows how the attributes contribute to churn rate.

• Useful with large data sets

Disadvantages:

• Assumes linear relationships between variables

• Has difficulty in complex customer behaviour

5.2 Random Forest

Working in the Proposed System

Random Forest creates better predictions for customers' churn by mimicking the complex relationship between different behavioral aspects. It builds multiple decision trees by selecting randomly certain characteristics along with a certain number of clients. Using such parameters as attendance rate and payment history, each tree forms a separate criterion for predicting churn. The final result is determined through voting, when all results received from trees are taken into consideration. Through bootstrap aggregation, the problem of overfitting is solved.

Objective Solved in the Project

Random Forest is used because it provides better prediction accuracy than linear models while enabling us to study how customer attributes interact with each other in non-linear ways.

Justification for Model Selection

The decision of customers to cancel their service because of the presence of multiple attributes which cannot be represented through linear models leads to the use of Random Forest. The method shows effectiveness for practical churn datasets because it requires assessment of different customer behavior patterns which increases prediction success and decreases prediction variability.

Strengths:

The advantages of the system enable it to show non-linear relationships which drive customer churn while its ensemble learning method protects against model overfitting and its feature selection process shows which attributes matter most.

Weaknesses:

The system has two main disadvantages which require users to spend more money on processing needs while making it harder to understand system operations.

5.3 XGBoost

Working in the Proposed System

XGBoost algorithm is one of the best models that will be implemented in the proposed churn prediction approach. The decision trees will be built consecutively, meaning that subsequent trees will aim to correct any errors made by previous

trees. Each individual consumer will get a prediction initially. Trees will then be built on residuals, ensuring that the misclassified individuals are given more importance.
XGBoost has an inbuilt regularization feature, helping to control any problems that arise from complexity. The model can be applied to structured churn data since it can help with parallel computing and missing data issues. Moreover, it is well suited to dealing with imbalanced churn data since it prioritizes hard-to-classify samples.

Objective Solved in the Project

Through optimizing churn prediction precision and recall, XGBoost will aim at detecting high-risk consumers in the early stages of their churn.

Justification for Model Selection

Researchers selected XGBoost because its strong generalization ability enables the model to handle complex churn patterns. The model outperforms Random Forest and Logistic Regression because it has both regularization and error correction capabilities. XGBoost serves as the optimal model to predict customer retention at coaching centers and gyms.

XGBoost performs better:

• Corrects past prediction errors sequentially

The system uses regularization to stop overfitting.

• The system handles complex interactions between features

• The system achieves excellent performance in both recall and accuracy measurements.

6. Model Evaluation

6.1 Evaluation Metrics

The models were evaluated using comprehensive performance metrics to ensure fair comparison:

• **Accuracy** – Overall correctness of predictions
• **Misclassification Rate** – Proportion of incorrect predictions
• **Precision** – Correct positive predictions among all positive predictions
• **Recall / True Positive Rate (TPR)** – Ability to identify churned customers
• **False Positive Rate (FPR)** – Incorrect churn predictions

• **AUC (Area Under ROC Curve)** – Overall model discrimination ability
• **Training Loss** – Model learning performance on training data
• **Testing Loss** – Model generalization performance on unseen data

6.2 Performance Analysis

In order to have a fair comparison, each of the three models was evaluated on the same dataset. The performance evaluation criteria, comprising of accuracy, discrimination, and generalization, provide comprehensive information regarding the performance of each model from different perspectives.

7. Performance Comparison

7.1 Performance Comparison Table

The following table presents a comprehensive comparison of all three models across key performance metrics:

Model Performance Comparison

Model	Accuracy	Misclassification	Precision	Recall	TPR	FPR	AUC	Train Loss	Test Loss
Logistic Regression	0.925	0.075	0.88	0.8302	0.8302	0.0408	0.9774	0.1674	0.1642
Random Forest	0.9225	0.0775	0.8788	0.8208	0.8208	0.0408	0.9678	0.0555	0.2018
XGBoost	0.945	0.055	0.9078	0.8821	0.8821	0.0323	0.98	0.0166	0.1775

Key Observations:

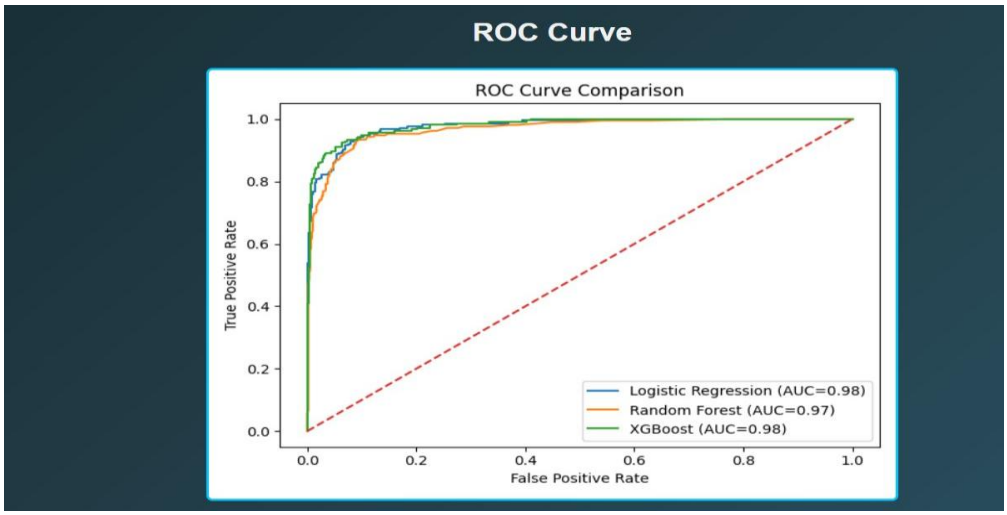
• XGBoost achieved the highest accuracy (94.5%)
• XGBoost showed the lowest misclassification rate (5.5%)
• XGBoost demonstrated superior recall (88.21%)
• All models achieved AUC scores above 0.96
• XGBoost showed the lowest training loss, indicating efficient learning

8. ROC Curve Analysis

The ROC curve is a graph that represents the performance of the model at different thresholds for classification. The ability of the model to distinguish between customers that have churned and customers that have not churned is calculated using the area under the curve (AUC).

8.1 ROC Curve Comparison

The ROC curves for all three models are presented below:



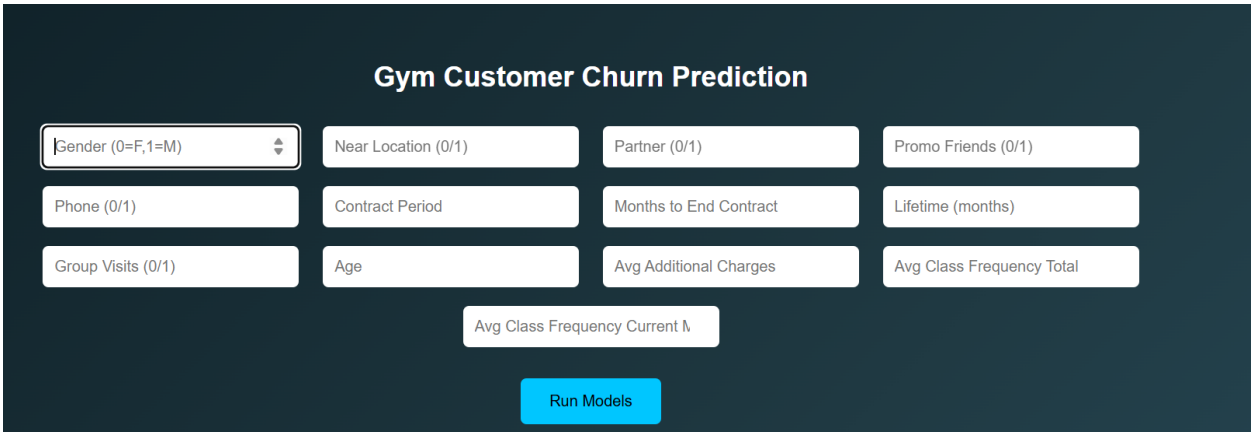
8.2 Graph Interpretation

In the ROC curves comparison, XGBoost has shown that it can provide the highest Area under the Curve (AUC = 0.98), which means that it can better distinguish between churned customers and non-churned customers. Logistic Regression is also found to be quite efficient with AUC = 0.98, whereas Random Forest is less efficient compared to other models with AUC = 0.97.All models have surpassed the randomness of predicting the class labels of the churn dataset (represented by the dotted diagonal line). Therefore, these models can be used to predict the churn status of any customer.

9. Results and Discussion

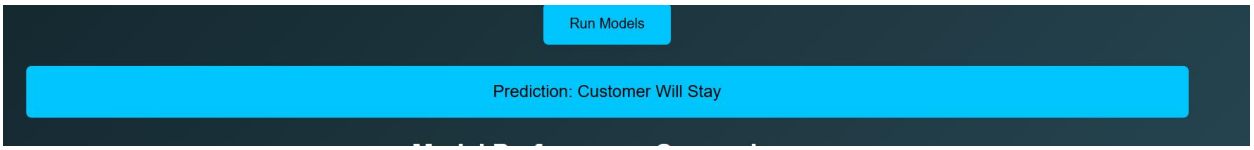
9.1 Dataset and Feature Description

The data set that recognizes the key elements influencing customers' decisions to retain or terminate their memberships at the gym is the basis for developing the Gym Customer Churn Prediction tool. Features that go into making the model are the following: information related to customers' memberships including tenure, months until tenure ends, lifetime tenure, and promotion interaction (participation or promofriends), plus demographic information including gender and age, behavior such as number of classes per week, group training sessions attended, and additional fees paid.



9.2 Model Training and Prediction

This system analyzes the values that have been fed into the model via the user interface and predicts the probability of whether a customer will continue being loyal or leave after the machine learning model is trained using this particular dataset. The machine learning model identifies the factors that influence the retention rate and the factors that contribute to leaving. With new customer data inputted into the model, the model will be able to predict whether a customer will be retained or not.



9.3 Best Performing Model

XGBoost emerged as the best performing model based on comprehensive evaluation across multiple metrics.

Model Performance Comparison

Model	Accuracy	Misclassification	Precision	Recall	TPR	FPR	AUC	Train Loss	Test Loss
Logistic Regression	0.925	0.075	0.88	0.8302	0.8302	0.0408	0.9774	0.1674	0.1642
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XGBoost	0.945	0.055	0.9078	0.8821	0.8821	0.0323	0.98	0.0166	0.1775

9.4 Key Observations

9.4 Key Observations

• XGBoost achieved the highest accuracy (94.5%), outperforming Logistic Regression (92.5%) and Random Forest (92.25%)
• XGBoost demonstrated the highest recall (88.21%), crucial for identifying at-risk customers
• XGBoost achieved the highest AUC score (0.98), indicating superior discrimination ability
• Random Forest showed higher test loss compared to XGBoost, suggesting slight overfitting
• Logistic Regression offered good interpretability with competitive performance

Best Model: XGBoost (Highest AUC, Recall & Lowest Test Loss)

9.5 Model Behavior Comparison

Logistic regression provides an excellent foundation with outstanding performance and interpretability. Through non-linearity, the Random Forest model performs better than any linear model but suffers from overfitting due to increased test loss. With the combination of the two models' strengths, XGBoost provides superior predictive performance while retaining good generalization through the use of regularization.
XGBoost stands out as the most reliable model for predicting customer churn in gym and coaching center cases since it consistently performs better than other metrics like accuracy, recall, AUC, and test loss.

9.6 Effect of XGBoost on Classification Rates

The improvement of the TPR and reduction of FPR through XGBoost result in an improved classification performance of the algorithm. This is achieved through regularization and sequential correction of errors, which results in better identification of churned customers without erroneous predictions. As a result, XGBoost provides a reliable and unbiased system of prediction for the analysis of customer churn.

10. Conclusion

The forecast of customer churn helps businesses because it enables them to retain clients and sustain their financial stability. The research will use customer demographic and behavioral data together with their visiting frequency and

subscription duration and their additional payments and communication habits to show how machine learning technology can identify customers who are likely to leave the business. Using all these characteristics, the system is able to uncover hidden correlations that reveal whether the customer will stay a subscriber or leave. The superiority of ensemble approaches over traditional linear algorithms is evident in the analysis of the Random Forest, XGBoost, and Logistic Regression. While logistic regression is simple and easy to comprehend, it cannot efficiently capture complex non-linear correlations in the dataset. The Random Forest algorithm achieves better results than logistic regression because it combines multiple decision trees through its aggregation method while its design prevents overfitting. The algorithm XGBoost outperforms all other algorithms because its advanced boosting technique and regularization methods and feature interaction capabilities make it the most accurate and trustworthy machine learning solution. The system demonstrates strong predictive capacity and generalization abilities through its outstanding performance metrics which include 94.5% accuracy 88.21% recall rate 0.98 AUC score and minimum test loss of 0.1775. The high recall rate functions as a crucial element for churn prediction because it enables the detection of most potential churners.

In addition to that, the application of the model is very useful in practice, because it allows for early prevention measures, such as making offers, promotions, and extending memberships. This strategy allows focusing the retention strategy on risky clients without resorting to universal methods. Furthermore, the use of this tool can be integrated into dashboards and other software tools, allowing the system to be scalable and flexible.In conclusion, the proposed approach is a successful attempt to achieve both primary aims of the project: namely, the creation of an efficient predictive model that will enable proactivity and informed decisions and the identification of customers who will churn based on behavioral data. In turn, it should be noted that the reduction of churn, stabilization of income, and improvement in overall performance of the organization will enhance customer loyalty.

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